

Deck Review — every claim checked against the build

Audited 2026-09-04 against commit d0c0422, branch main, working tree clean.

Method Each slide claim traced to a file in the repository, or recorded as absent. Figures read from ui/payload.json, out/fixture.manifest.json and the test suite.

Result 13 claims verified · 3 factually wrong about our own stack · 5 roadmap items presented as shipped · 1 limitation slide to add.

The deck is close. Its problem is not exaggeration in the usual direction — it is that three slides describe a *different* system than the one we built, and in two of those cases the thing we actually built is the better story. Fixing them costs four edits and makes the deck stronger, not weaker.

01 Verified — shipped, measured, openable on demand

Each row is a deck claim that survives contact with the repository. The right-hand column is what a judge sees if they ask us to run it.

DECK CLAIM	WHERE IT LIVES	MEASURED
Adversarial Verification Loop	core/verify/ - one binary	33 admitted before the wipe, 0 admitted after. Candidates scanned 58 → 39. Parameter identity between the two carves is enforced by construction, not by convention.
Sanitize Behavioural Audit	core/wipe/src/audit.rs	A simulated ATA SANITIZE returned in 1,000 ns against a floor of 431,059,458 ns — 431,059.458× faster than physically possible. Verdict UNVERIFIED_TIMING.
Corroboration of that verdict	wipe witness block	256 sampled sectors hashed before and after: unchanged: true . The drive did nothing, and two independent methods say so.
Media-appropriate method	--method, --sanitize	3 overwrite methods, 6 firmware primitives. method_selected_by: detected-medium — never one-size-fits-all.
Confidence score per file	core/carve/src/confidence.rs	Published, not asserted: 0.40·sig + 0.35·struct + 0.15·entropy + 0.10·size , admission gate 0.75 .
Structure-aware carving	7 validators: jpeg png pdf zip sqlite mp4 gzip	28 of 40 recovered by the default engine, 30 of 40 with reassembly, against a reachability ceiling of 33 of 40 . Asserted as a test, not observed.
Fragmented-file reassembly	core/carve/src/bifragment.rs	5 planted files require it. Bounded two-extent search over split point × gap, quantised to the cluster grid. Off by default and it says so.
Signed certificate	core/ledger/ - Ed25519, RFC 8785	Signature covers the <i>entire</i> certificate, both regions: 1,622 canonical bytes . No float ever enters the signed payload.
Tamper-evidence	tests/test_ledger_tamper.py	One flipped byte fails verification and the offending field is named. Merkle chain head anchors ordering.

DECK CLAIM	WHERE IT LIVES	MEASURED
Stating verification limits	<code>outcome.whole_medium_claim</code>	false , with coverage 0.001953 printed on the certificate face rather than in a footnote.
Crypto-Erase Shim	<code>--crypto-erase-demo</code>	Real, at 512-byte block granularity, key destroyed on drop — and labelled a <i>demonstration</i> in every field it emits. See §04.
Provenance tagging	<code>carve.records[].assembly</code>	Every record states how its bytes were obtained: contiguous, reassembled, or signature-span . Observed versus derived, per record.
Zero hardware gatekeeping	image files, no vendor tooling	True. Nothing is mounted; the carver reads bytes, so the image is a file and never an attached device.

02 Wrong about our own stack — fix before this ships

These are the dangerous ones. Not because they overclaim, but because a judge who opens the repository finds the contradiction in under a minute, and a contradiction found in the deck makes every other number in it a question.

Slide 3 says the desktop app is Electron FACTUALLY WRONG

There are **zero** Electron references in the tree. We ship **Tauri v2**. This is not pedantry: the Tauri choice is a selling point we are currently throwing away and replacing with the thing every other team says.

SLIDE 3, LEFT COLUMN

~~DESKTOP APP / Electron~~

DESKTOP APP / Tauri v2 – 4.8 MB shell over the platform's own webview, no bundled Chromium, offline by construction

Slide 3 says “Hyperledger Fabric commit” NOT BUILT

We ship a local Ed25519 signature over RFC 8785 canonical bytes plus a hand-rolled Merkle chain. No Fabric anywhere. Our own scope document puts Fabric out until nationals — so the deck is currently promising something we have already decided not to build yet.

The replacement is the stronger claim anyway: roughly 120 lines we can explain under questioning beats a framework we would have to hand-wave.

SLIDE 3, VERIFY & ANCHOR

~~SHA-256 hash, timestamp, Hyperledger Fabric commit~~

SHA-256 leaf, Ed25519 signature over RFC 8785 canonical bytes, local Merkle chain – Fabric anchoring at nationals

Slide 3 says “Tamper-evident PDF and JSON” HALF ABSENT

There is no PDF emitter in the repository. The certificate is JSON, and its tamper-evidence is real — but naming an artifact we do not produce invites the one demo request we cannot satisfy.

SLIDE 3, SIGNED CERTIFICATE

~~Tamper-evident PDF and JSON, one per job~~

Tamper-evident signed JSON certificate, one per job – one flipped byte fails verification and names the field

03 Roadmap presented as shipped — keep, but mark

None of these need to leave the deck. Vision belongs in an idea submission. But each currently reads as a working feature, and the fix is a single word — a label, a colour, a “Phase 7” marker — not a deletion.

“40+ hidden data ghosts across journals, caches, and AI vector stores” NOT BUILT

Not implemented. The fixture's 40 planted files are ordinary documents — txt, gzip, png, jpeg, pdf, docx, sqlite, mp4. No journals, no caches, no vector stores. Mark it as the roadmap item it is.

File & Folder Eraser — slack space, shadow copies, ADS, NTFS / ext4 / exFAT / APFS NOT BUILT

None of it exists. We perform whole-image overwrite. The string exFAT appears nowhere in the tree, and APFS appears only in documents that list it as out of scope.

ML carving PHASE 7

Not built, and correctly identified as a stretch goal in our own scope discipline. Signature and structure carving are what ship. Saying so is fine; the arrow diagram on slide 3 currently implies all four stages run today.

“RR-score” METRIC DOES NOT EXIST

Slide 4 claims we “measure real adversary recovery failure (RR-score).” The *measurement* is real and is the best thing we have — 33 admitted before, 0 after. The *acronym* exists nowhere in the codebase. An invented metric name is precisely what an NTRO panel probes, and it would be the one question we could not answer.

SLIDE 4, PHYSICALLY HONEST

~~we measure real adversary recovery failure (RR score)~~

we attack our own output with our own carver and publish the result: 33 candidates admitted before the wipe, 0 after

“Runs 100% in software via native block I/O & Sleuth Kit” UNDERSELLS US

This reads as *we wrapped The Sleuth Kit*. We did not. We implemented carving ourselves; TSK is an independent cross-check we ran file by file and documented in docs/tsk_crosscheck.md, comparing by SHA-256 rather than row count. The current phrasing gives away our strongest engineering claim to a third-party library.

SLIDE 2, SOFTWARE-ONLY CONSTRAINT

~~Runs 100% in software via native block I/O & Sleuth Kit — zero hardware dependencies~~

Runs 100% in software via native block I/O. Carving is our own engine; The Sleuth Kit is the independent cross-check we validate against, file by file, by SHA-256

04 Numbers to add, slide by slide

Every figure below traces to a measurement in the repository. Nothing here is illustrative. Slides 2 and 5 currently carry no numbers at all, which is the single largest missed opportunity in the deck: our whole argument is that we measure what others assert.

Slide 2 — Our Innovation: five slogans, zero figures

ROW	ADD
Adversarial Verification Loop	33 admitted → 0 admitted. Same carver, same parameters, enforced by construction.

ROW	ADD
Sanitize Behavioural Audit	431,059× faster than physically possible — and the drive returned success.
Shadow Graph Traversal	Roadmap. No figure — do not invent one.
Crypto-Erase Shim	512-byte block granularity, key destroyed on drop, labelled a demonstration in every field it emits.
Provenance-Tagged Recovery	Every record states how its bytes were obtained: contiguous, reassembled, signature-span .

Slide 3 — Technical Approach: capability lists with no throughput

- **268,435,456 bytes** overwritten in **435,684,084 ns** — **616,124,081 B/s**.
- Live sector telemetry at **110.17 Hz** measured against a declared **20 Hz** floor; longest uninstrumented interval **8.674 ms**.
- Whole loop — carve, wipe, carve again, sign, chain — **7.32 s** end to end on the demo laptop.

Slide 4 — Feasibility: replace the invented acronym with the real pair

- Whole-image entropy **7.061690** → **7.999999** bits per byte, same estimator both sides, over all **268,435,456** bytes.
- Admission margin: lowest admitted **0.90**, highest rejected **0.65** — a population gap of **0.25** against a gate at **0.75**.
The gate is not sitting on top of the data.

Slide 5 — Impact: three prose boxes, not one number

This is the weakest slide in the deck. It makes three large claims and offers no evidence that anything was built. Put the engineering underneath them:

- **548** Rust tests and **13** Python suites.
- A **1.1 MB** engine binary and a **4.8 MB** desktop shell — offline by construction, no Chromium.
- Ground truth is a signed manifest: **40** planted files of known SHA-256, of which **28** are contiguous, **5** require reassembly, and **7** we made unrecoverable deliberately, to defeat our own engine.

05 The limitation slide

Add one. It should be this one, and the copy below is paste-ready. It is the right choice for three reasons: it is already printed on the signed certificate, so we are not inventing a slide; it pre-empts the hardest question a forensics officer can ask; and it arrives with a stated remedy and a stated cost.

PASTE-READY · NEW SLIDE, OR THE LOWER THIRD OF SLIDE 4

What we do not claim: sampled verification is not proof of the whole medium

This run read back **1,024 of 524,288 sectors — 0.1953% coverage**. The longest run of consecutive sectors the sampling plan never touched is **2,394 sectors (1,225,728 bytes)**. A region left unwiped inside that window would pass our own verdict.

`--verify exhaustive` closes the gap by reading every sector. It is not the default, because a default that takes minutes is a default nobody runs.

We publish this on the face of the certificate, not in a footnote. NIST SP 800-88 advises stating verification limits rather than asserting a clean wipe. A tool that hid this number would be claiming something it did not measure.

Runner-up, if a second limitation is wanted

The signature proves integrity, not authority

Key custody is none — generated locally, unattested. A forged certificate is detectable by anyone holding the public half, which is a real and useful property. It does not make the document *attributable*. The production path is an operator key enrolled against an organisational CA or HSM. This too is already on the certificate's own face.

06 Two internal inconsistencies to close first

Neither is in the deck, but both are one grep away from a judge who reads our documents, and both undercut the claim the deck is built on.

docs/standards_map.md is stale on signing

It still reads “The certificate is not yet signed. Ed25519 and Merkle anchoring are Phase 4.” Phase 4 landed in commit `bc2b838`. The document that an officer reads before the code currently says we have not built the thing the deck's strongest slide is about.

The same file quotes the largest unsampled run as 2,815 sectors where the current payload says 2,394. Both are legitimate per-run values — the sampling plan is seeded — but the document presents its figure as fixed, and a reader comparing the two will assume one is wrong.

loop.recall_before: 28 is a literal, not a reading

It is hard-coded at `ui/build_payload.py:139`. The number itself is sound — `tests/test_carve_recall.py:217` asserts it against a real carve run — but the payload does not derive it. If anyone asks where 28 comes from, the honest answer is the test suite, not the file on screen.

Correction to my earlier note: I suggested “35 of 35 admitted records byte-matched ground truth” for slide 2. Do not use it. That figure comes from `fixtures/sample_output.json`, whose own provenance block records `is_carve_run: false` — it is a frozen generator sample, not a carve. The measured pair to put on a slide is 33 admitted → 0 admitted.

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Sources: ui/payload.json · out/fixture.manifest.json · docs/standards_map.md · docs/tsk_crosscheck.md ·
tests/test_carve_recall.py · core/{carve,wipe,ledger,verify,device}

Every figure in this document was read from a committed artifact or a passing test. No value here is illustrative.